

Econ 30065: Microeconomics

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Consumer's Choice

Building blocks of the Consumer's Problem

1. Preferences

- What people like

2. Budget Constraints

- What people can afford

3. Choice

4. From choice to demand functions

“Affordable bundles”: budget set

- A bundle of goods (X,Y) is affordable if:
It costs less than the amount of money the consumer has to spend (income M):

$$P_X X + P_Y Y \leq M$$

- This is the consumer's ***budget constraint***
- The *budget set* is the set containing all the affordable bundles
- The *budget line* contains all the bundles that use all the income M:

$$P_X X + P_Y Y = M$$

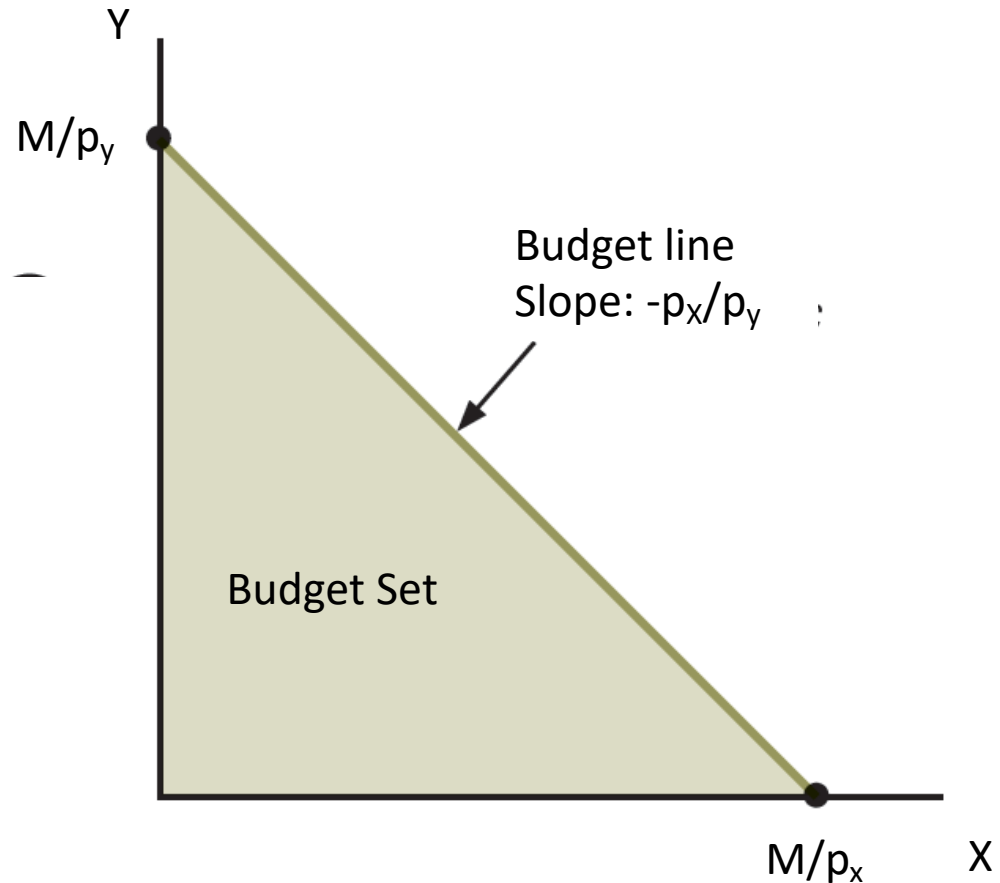
Budget Set

➤ Rearranging the budget line equation:

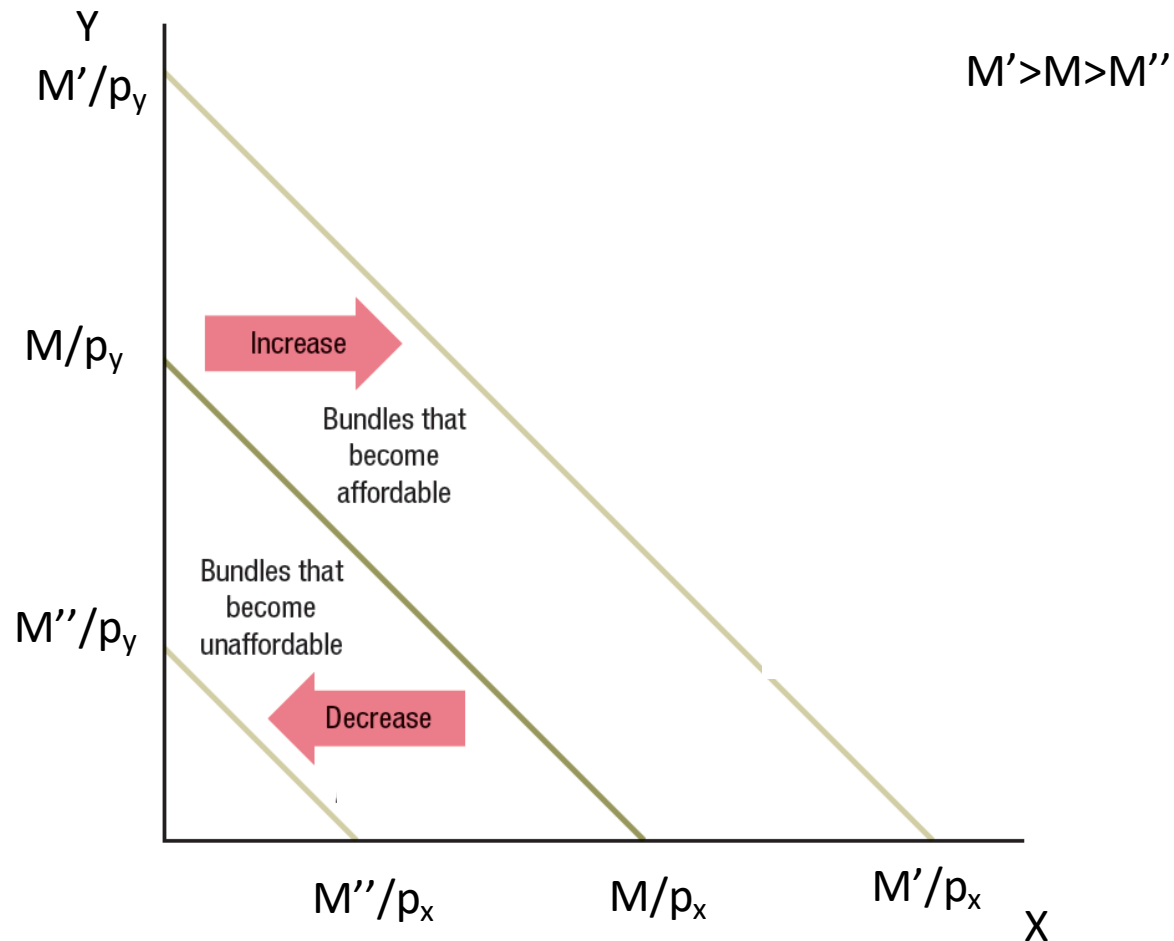
$$Y = \frac{M}{P_Y} - \frac{P_X}{P_Y} X$$

➤ Bundles in the shaded area are affordable but do not exhaust income

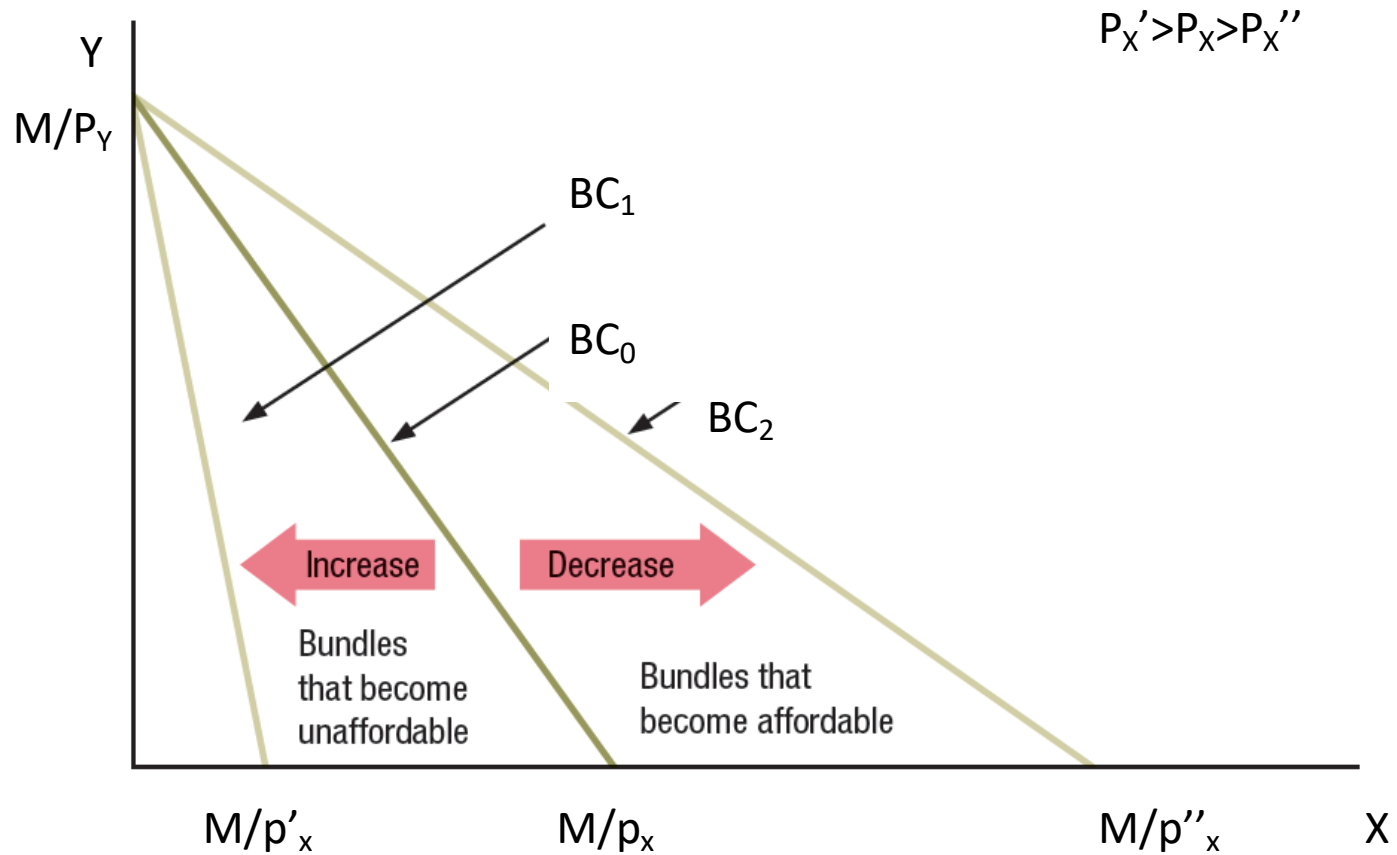
➤ Bundles on the budget line exhaust income



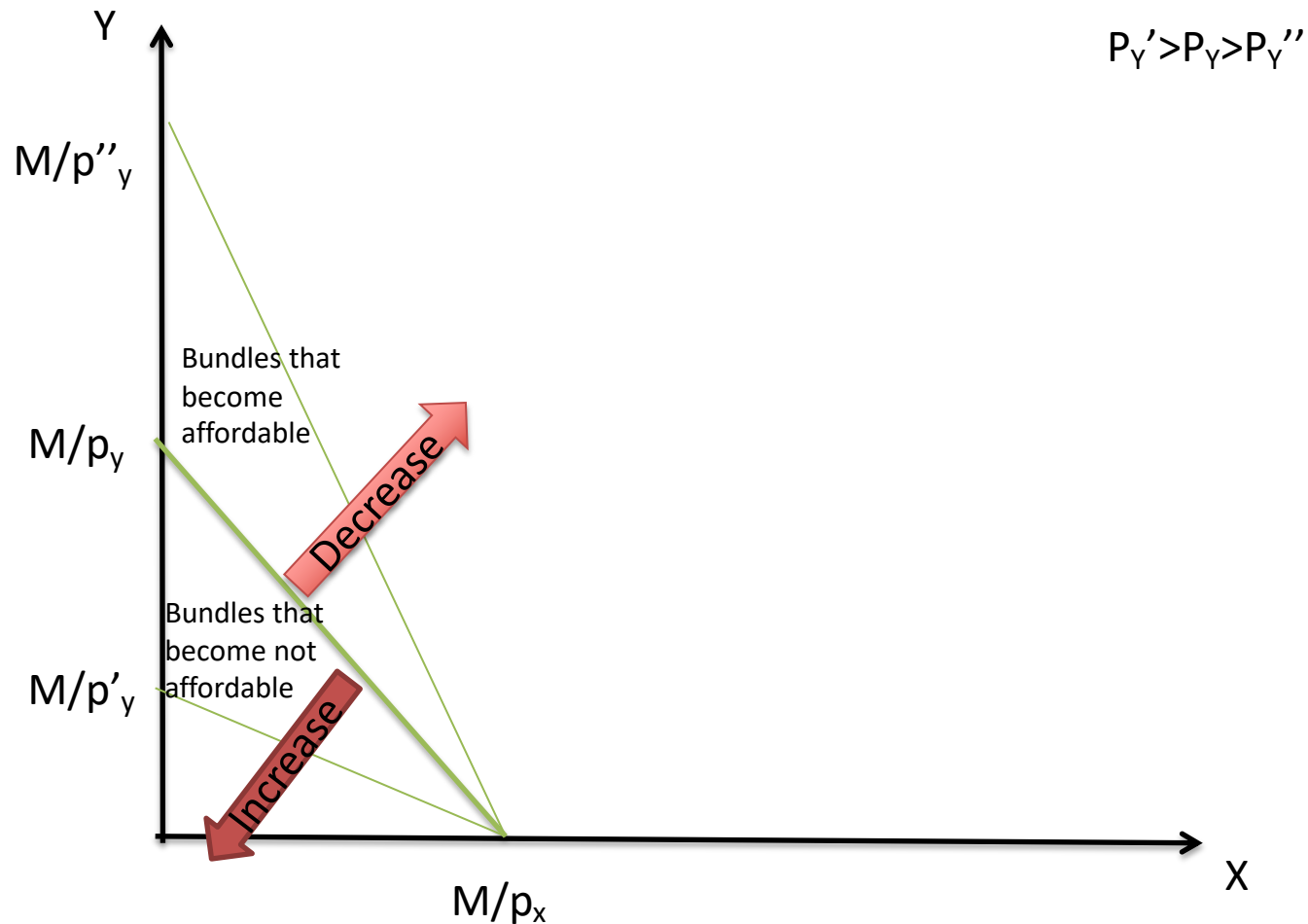
Changes in Income and the Budget line



Changes in Price P_X and the Budget line



Changes in Price P_Y and the Budget line



Summary: Properties of Budget line

- Budget line is the boundary that separates affordable bundles from all others
- Slope of budget line = $-P_X/P_Y$
- X-intercept is M/P_X ; Y-intercept is M/P_Y
- Change in income shifts the line without changing its slope
- Change in the price of a good rotates the line

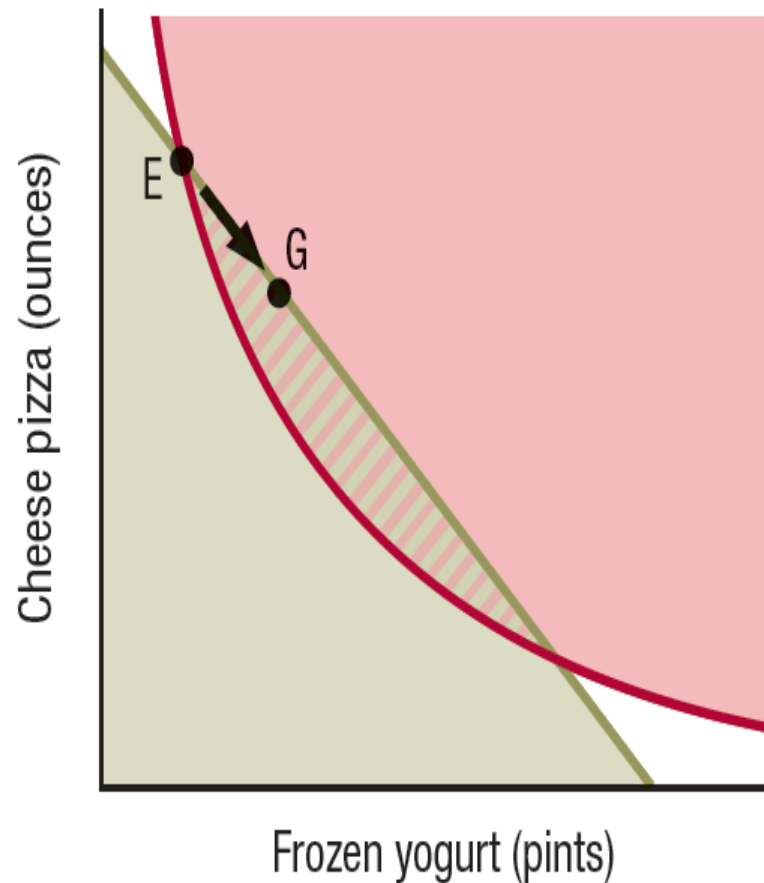
Consumer Choice

- Choice principle suggests a consumer will choose the highest-ranked available option
- Graphically:
 - A bundle on the budget line, not below it
 - A bundle on the highest indifference curve that touches the budget line
 - tangency condition
- Mathematically: $\max U(X,Y)$ given budget set

Choosing among affordable bundles

- E is not a best choice because there are other affordable bundles that are better than E (for example, G)

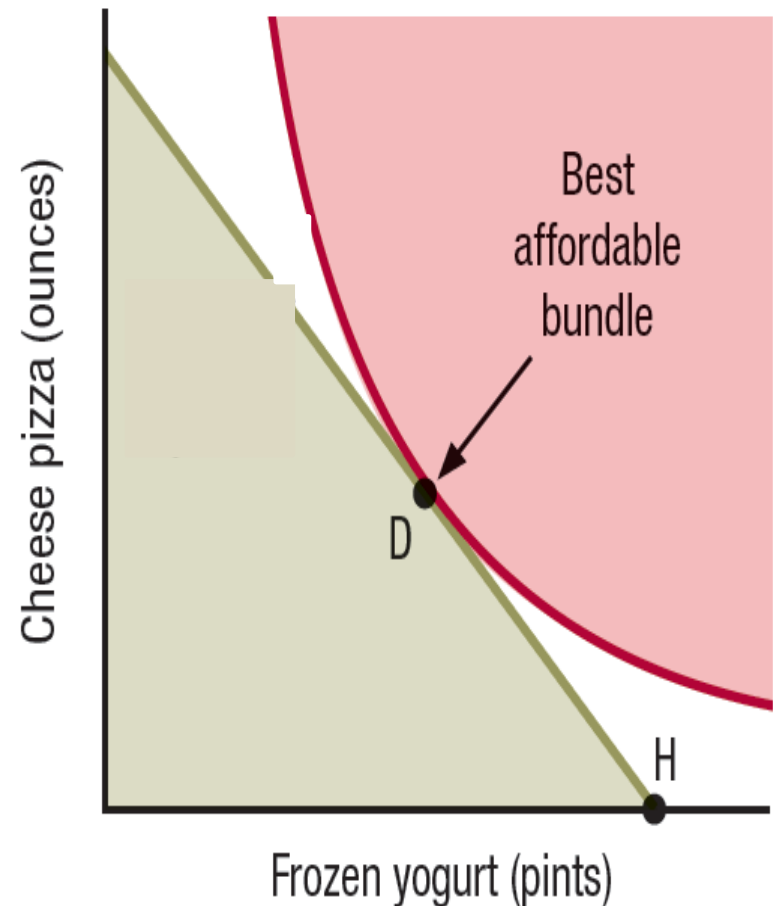
(b) Not a best choice



Choosing among affordable bundles

D is the best choice
because no affordable
bundle lies on a higher
indifference curve

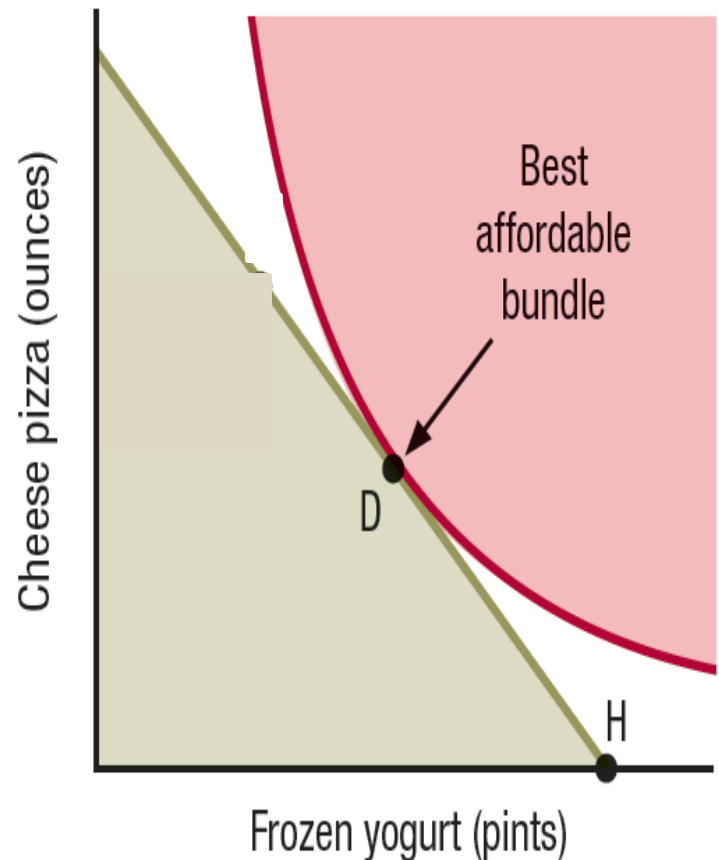
(a) A best choice



Interior solutions

- At every **interior solution**, the budget line lies tangent to the indifference curve at the chosen consumption bundle
- Recall that:
 - Slope of the indifference curve is $-MRS_{XY}$
 - And slope of the budget line is $-P_X/P_Y$
- Thus at an **interior solution**:
 - $MRS_{XY} = P_X/P_Y$
- Optimal bundle must lie on the budget line:
 - $P_X X + P_Y Y = M$

(a) A best choice



Interior solution math procedure

- Max $U(X,Y)$, given $P_X X + P_Y Y = M$
- To find the optimal bundle solve the system of equations:

$$MRS_{xy} = P_X / P_Y$$

$$P_X X + P_Y Y = M$$

Example of interior solutions

- Lauren spends part of her income on pet food for her cat Susy. Her utility function is given by ,

$$U(X, Y) = XY^4$$

- where x is the number of pet food tins and y is a composite of “Other goods”.
- Assuming that Lauren’s monthly income is 300 € and that the prices of x and y are, respectively, $p_x = 0.4$ € and $p_y = 1$ €, find analytically her optimal bundle.

Analytical solution

➤ The marginal rate of substitution is $MRS = (1/4)(y/x)$.

➤ The price ratio is: 0.4

➤ Thus:

$$\begin{cases} \frac{y}{4x} = 0.4 \\ 0.4x + y = 300 \end{cases} \Rightarrow \begin{cases} x = 150 \\ y = 240 \end{cases}$$